MH1100 Calculus I

Handout 12 - Inverse Function

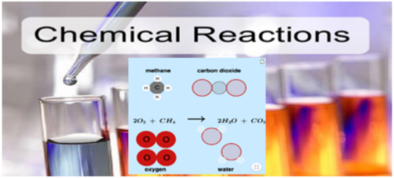
Synopsis

Inverse Function

- Inverse Functions
- Exponential Functions and Their Derivatives
- Logarithmic Functions
- Derivatives of Logarithmic Functions

Further Reading:

Euler's formula; Euler's identity; Harmonic series; Complex analysis; Normal distribution; Logistic function; Logistic regression; Diffusion equation; Brown motion; Exponential decay; Half-life; Average lifetime; ...

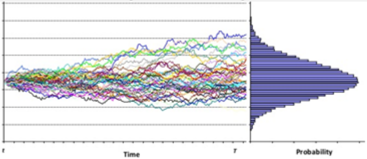


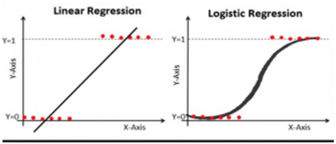
Euler's Formula

$$e^{i\phi} = \cos\phi + i\sin\phi$$

Euler's identity

$$e^{i\pi} + 1 = 0$$

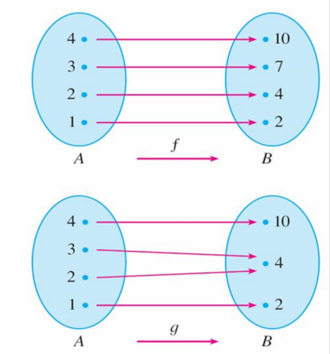




Inverse Functions; Exponential Functions and Their Derivatives; Logarithmic Functions; Derivatives of Logarithmic Functions

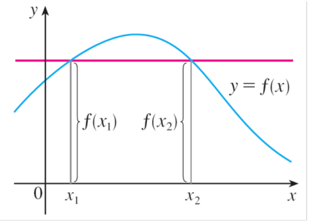
1 Definition A function f is called a **one-to-one function** if it never takes on the same value twice; that is,

$$f(x_1) \neq f(x_2) \qquad \text{whenever } x_1 \neq x_2$$



f is one-to-one; g is not.

If a horizontal line intersects the graph of f in more than one point, then there exist two numbers x_1 and x_2 such that $f(x_1) = f(x_2)$.



This function is not one-to-one because $f(x_1) = f(x_2)$.

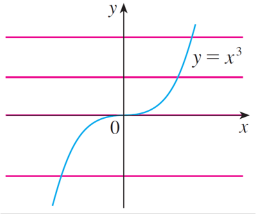
This means that f is not one-to-one. Therefore we have the following geometric method for determining whether a function is one-to-one.

> **Horizontal Line Test** A function is one-to-one if and only if no horizontal line intersects its graph more than once.

Example

Is the function $f(x) = x^3$ one-to-one?

Solution 1:
If $x_1 \neq x_2$, then $x_1^3 \neq x_2^3$ (two different numbers can't have the same cube). Therefore, by the Horizontal Line Test, $f(x) = x^3$ is one-to-one.
Solution 2: From the figure of function x^3, we can see that no horizontal line intersects the graph of $f(x) = x^3$ more than once. Therefore, $f(x)$ is one-to-one.



$f(x) = x^3$ is one-to-one.

Inverse Functions

2 Definition Let f be a one-to-one function with domain A and range B. Then its **inverse function** f^{-1} has domain B and range A and is defined by

$$\underbrace{f^{-1}(y)}_{y \in B} = \underbrace{x}_{x \in A} \iff f(x) = y \qquad \frac{4y^2}{y^2+1} = x$$

for any y in B.

The letter x is traditionally used as the independent variable, so when we consider f^{-1}, we reverse the roles of x and y

3

$$\underbrace{f^{-1}(x)}_{x \in B} = \underbrace{y}_{y \in A} \iff f(y) = x$$

We also have the following **cancellation equations**:

4

$$f^{-1}(f(x)) = x \quad \text{for every } x \text{ in } A$$

$$f(f^{-1}(x)) = x \quad \text{for every } x \text{ in } B$$

Compute inverse functions!

If we have a function $y = f(x)$ and are able to solve this equation for x in terms of y, then we must have $x = f^{-1}(y)$.

If we want to use symbol (or notation) x as the independent variable, we then interchange x and y and arrive at the equation $y = f^{-1}(x)$.

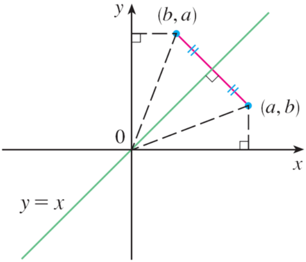
5 How to Find the Inverse Function of a One-to-One Function f

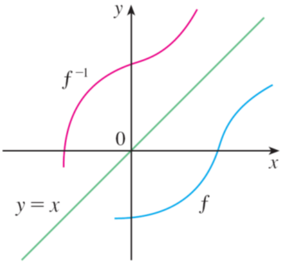
STEP 1 Write $y = f(x)$.

STEP 2 Solve this equation for x in terms of y (if possible). $x = \ldots$

STEP 3 To express f^{-1} as a function of x, interchange x and y. The resulting equation is $y = f^{-1}(x)$.

Geometrically, since $f(a) = b$ if and only if $f^{-1}(b) = a$, the point (a, b) is on the graph of $f(x)$ if and only if the point (b, a) is on the graph of $f^{-1}(x)$.





We get the point (b, a) from (a, b) by reflecting about the line $y = x$ as shown in figure.

> The graph of f^{-1} is obtained by reflecting the graph of f about the line $y = x$.

Example

Find the inverse function $f(x) = x^3 + 2$.

Solution

We have

$$y = x^3 + 2.$$

We can solve the equation for x,

$$x^3 = y - 2$$

$$x = \sqrt[3]{y - 2}$$

Then we interchange x and y, we have

$$y = \sqrt[3]{x - 2}$$

$y=x^2+2$ will not have inverse function in the region $[-2, 2]$, but have for the region $[0,2]$.

Inverse Functions

6 Theorem If f is a one-to-one continuous function defined on an interval, then its inverse function f^{-1} is also continuous.

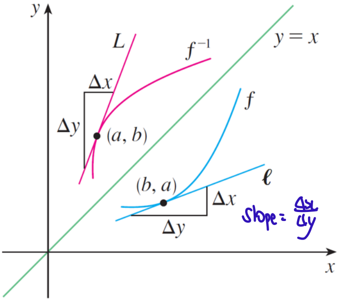
7 Theorem If f is a one-to-one differentiable function with inverse function f^{-1} and $f'(f^{-1}(a)) \neq 0$, then the inverse function is differentiable at a and

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}$$

We get the graph of f^{-1} by reflecting the graph of f about the line $y = x$, so the graph of f^{-1} has no corner or kink in it either.

We therefore expect that f^{-1} is also differentiable (except where its tangents are vertical). In fact, we can predict the value of the derivative of f^{-1} at a given point geometrically.

The graphs of f and its inverse f^{-1} is demonstrated below.



$f(b) = a \Rightarrow f^{-1}(a) = b$

Inverse Functions

If $f(b) = a$, then $f^{-1}(a) = b$ and $(f^{-1})'(a)$ is the slope of the tangent line L to the graph of f^{-1} at (a, b), which is $\Delta y/\Delta x$.

Reflecting in the line $y = x$ has the effect of interchanging the x- and y-coordinates. So the slope of the tangent line ℓ (of the reflected curve) at point (b, a) is $\cancel{\Delta y/\Delta x}$. $\Delta x/\Delta y$

Thus the slope of L is the reciprocal of the slope of ℓ, that is,

$$(f^{-1})'(a) = \frac{\Delta y}{\Delta x} = \frac{1}{\Delta x/\Delta y} = \frac{1}{f'(b)}$$

$y = f(x)$ $\quad$ $x = f^{-1}(y)$

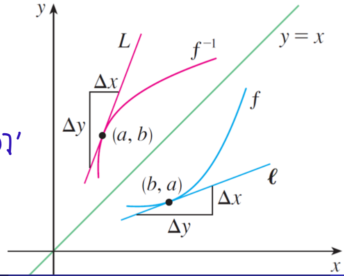
$\downarrow$ $\quad$ $\downarrow$

$dy = f'(x)\,dx$ $\quad$ $dx = [f^{-1}(y)]'\,dy$

$\downarrow$ $\quad$ $\downarrow$

$\frac{dx}{dy} = \frac{1}{f'(x)}$ $\quad$ $\frac{dx}{dy} = [f^{-1}(y)]'$

$= \frac{1}{f'(f^{-1}(y))}$ equivalent



Inverse Functions

inverse function exists → continuous

7 Theorem If f is a one-to-one differentiable function with inverse function f^{-1} and $f'(f^{-1}(a)) \neq 0$, then the inverse function is differentiable at a and

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}$$

continuous ∵ original function is differentiable

Proof

We let $f(b) = a$, so $f^{-1}(a) = b$. We let $y = f^{-1}(x)$, then $f(y) = x$. The derivative $(f^{-1})'(a)$ can be evaluated by

$$(f^{-1})'(a) = \lim_{x \to a} \frac{f^{-1}(x) - f^{-1}(a)}{x - a} = \lim_{y \to b} \frac{y - b}{f(y) - f(b)}$$

$$= \lim_{y \to b} \frac{1}{\frac{f(y)-f(b)}{y-b}} = \frac{1}{\lim_{y \to b} \frac{f(y)-f(b)}{y-b}} = \frac{1}{f'(b)} = \frac{1}{f'(f^{-1}(a))}$$

Example

The function $f(x) = x^2$, $0 \leq x \leq 2$, is a one-to-one function. Find $(f^{-1})'(1)$. $a=1$

Solution

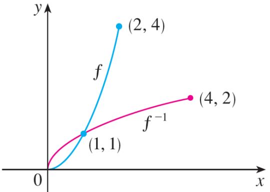
domain does not allow for -1.

Since $f(1) = 1$, we have $f^{-1}(1) = 1$. Further $f'(x) = 2x$, thus we have

$$(f^{-1})'(1) = \frac{1}{f'(f^{-1}(1))} = \frac{1}{f'(1)} = \frac{1}{2}$$

In fact, $f^{-1}(x) = \sqrt{x}$, $0 \leq x \leq 4$. Then $(f^{-1})'(x) = 1/(2\sqrt{x})$, so $(f^{-1})'(1) = \frac{1}{2}$, which agrees with the preceding computation.

$y = x^2$
$x = \sqrt{y}$
$y = \sqrt{x}$
$y' = \frac{1}{2\sqrt{x}}$



Inverse Functions

Example

If $f(x) = 2x + \cos x$, find $(f^{-1})'(1)$.

① $f'(x)$

② $f^{-1}(a)$

$a=1$

Solution

Note that $f(x)$ is one-to-one, because

$$f'(x) = 2 - \sin x > 0$$

$f^{-1}(1)$

$2x + \cos x = 1$

$\Rightarrow$ sometimes hard to find a value for x. may have to do guesses and use Newton's method.

so $f(x)$ is increasing. Further we have

$$f(0) = 1 \Rightarrow f^{-1}(1) = 0$$

find by trial and error.

Therefore

$$(f^{-1})'(1) = \frac{1}{f'(f^{-1}(1))} = \frac{1}{f'(0)} = \frac{1}{2 - \sin 0} = \frac{1}{2}$$

Understanding Checker

Q1. We have (a) $\ln(x+\sqrt{x^2+1})$; (b) $\ln\left(\frac{1-x}{1+x}\right)$. They are even or odd functions?

(A) (a) even (b) even (B) (a) even (b) odd (C) (a) odd (b) even (D) (a) odd (b) odd

even: $f(-x)=f(x)$ odd: $f(-x)=-f(x)$

$\ln(-x+\sqrt{x^2+1})$
$=\ln\frac{1}{x+\sqrt{x^2+1}}$
$=-\ln(x+\sqrt{x^2+1})$
$=-f(x)$
$\therefore$ odd function

$\ln\left(\frac{1-(-x)}{1+(-x)}\right)$
$=\ln\left(\frac{1+x}{1-x}\right)$
$=\ln\left(\frac{1-x}{1+x}\right)^{-1}$
$=-\ln\left(\frac{1-x}{1+x}\right)$
$=-f(x)$
$\therefore$ odd function

Q2. We have $f'(x)=2x-3\sin x$ and $f(0)=5$, then $f(x)$ is

(A) $x^2-3\cos x+8$ (B) $x^2+3\cos x+5$ (C) $x^2+3\cos x+2$ (D) $\frac{x^2}{2}+3\cos x+2$

$f(x)=2\frac{x^2}{2}-3(-\cos x)+C$

$=x^2+3\cos x+C$

$f(0)=5 \Rightarrow C=2$

$\therefore f(x)=x^2+3\cos x+2$

C

Q3. For function $f(x)=x+\cos x$, use Newton's method, if we choose $x_1=\frac{\pi}{2}$, then we have x_2 equal to

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) 0 (D) D.N.E

$f(\frac{\pi}{2})=\frac{\pi}{2}+\cos\frac{\pi}{2}=\frac{\pi}{2}$

$X_{n+1}=X_n-\frac{f(x_n)}{f'(x_n)}$

$f'(x)=1-\sin x$

$x_2=x_1-\frac{f(x_1)}{f'(x_1)}$

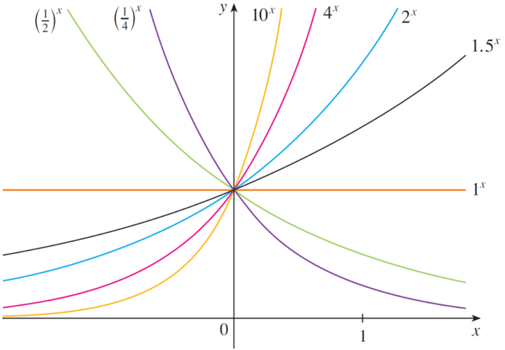
$=\frac{\pi}{2}-\frac{\frac{\pi}{2}+\cos\frac{\pi}{2}}{1-\sin\frac{\pi}{2}}$ $=0$ $\therefore$ not well defined

$\therefore$ D

Exponential Functions and Their Derivatives

The graphs of **exponential function** $y = b^x$ are shown in figure below for various values of the base b.

$x = 1$
$y = b^1 = b$



Member of the family of exponential functions

Exponential Functions and Their Derivatives

All exponential function graphs pass through the same point $(0, 1)$, because $b^0 = 1$ for $b \neq 0$.

As the base b gets larger, the exponential function grows more rapidly for $(x > 0)$.

grows much faster as $x \to \infty$

$\log_b x \ll x^p \ll b^x \quad (b>1, p>0)$

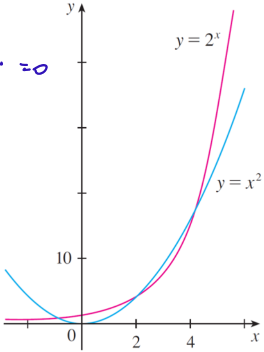
$x \to \infty$

$\lim_{x\to\infty} \frac{\log_b x}{x^p} = 0$

$\lim_{x\to\infty} \frac{x^p}{b^x} = 0$

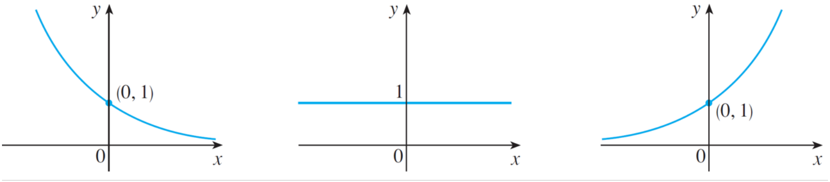
Compare exponential function $y = 2^x$ with the power function $y = x^2$.

The graphs intersect three times, but ultimately the exponential curve $y = 2^x$ grows far more rapidly than the parabola $y = x^2$.



Exponential Functions and Their Derivatives

There are basically three kinds of exponential functions $y = b^x$.
If $0 < b < 1$, the exponential function decreases;
If $b = 1$, it is a constant;
If $b > 1$, it increases.



$y = b^x$, $0 < b < 1$ $\quad$ $y = 1^x$ $\quad$ $y = b^x$, $b > 1$

Because $y = (1/b)^x = 1/b^x = b^{-x}$, the graph of $y = (1/b)^x$ is just the reflection of the graph of $y = b^x$ about the y-axis.

Exponential Functions and Their Derivatives

2 Theorem If $b > 0$ and $b \neq 1$, then $f(x) = b^x$ is a continuous function with domain $\mathbb{R}$ and range $(0, \infty)$. In particular, $b^x > 0$ for all x. If $0 < b < 1$, $f(x) = b^x$ is a decreasing function; if $b > 1$, f is an increasing function. If $a, b > 0$ and $x, y \in \mathbb{R}$, then

1. $b^{x+y} = b^x b^y$ **2.** $b^{x-y} = \dfrac{b^x}{b^y}$ **3.** $(b^x)^y = b^{xy}$ **4.** $(ab)^x = a^x b^x$

3 If $b > 1$, then $\lim_{x\to\infty} b^x = \infty$ and $\lim_{x\to-\infty} b^x = 0$

If $0 < b < 1$, then $\lim_{x\to\infty} b^x = 0$ and $\lim_{x\to-\infty} b^x = \infty$

$y=0$ is horizontal asymptote

In particular, if $b \neq 1$, then the x-axis is a horizontal asymptote of the graph of the exponential function $y = b^x$.

Exponential Functions and Their Derivatives

Example

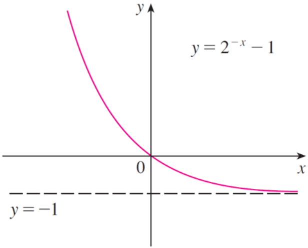
(a) Find $\lim_{x\to\infty}(2^{-x} - 1)$.
(b) Sketch the graph of the function $y = 2^{-x} - 1$.

Solution

$0 < b < 1$

(a) $\lim_{x\to\infty}(2^{-x} - 1) = \lim_{x\to\infty} \left[(\frac{1}{2})^x - 1)\right] = 0 - 1 = -1$.

(b) We write $y = [(\frac{1}{2})^x - 1]$ as in part (a). The graph is shown in figure below. The line $y = -1$ is the horizontal asymptote.



Exponential Functions and Their Derivatives

10 Properties of the Natural Exponential Function The exponential function $f(x) = e^x$ is an increasing continuous function with domain $\mathbb{R}$ and range $(0, \infty)$. Thus $e^x > 0$ for all x. Also

$e \approx 2.71818...$

$$\lim_{x \to -\infty} e^x = 0 \qquad \lim_{x \to \infty} e^x = \infty$$

So the x-axis is a horizontal asymptote of $f(x) = e^x$.

For example, if we want to calculate $\lim_{x\to\infty} \frac{e^{2x}}{e^{2x}+1}$

We can divide numerator and denominator by e^{2x}:

$$\lim_{x\to\infty} \frac{e^{2x}}{e^{2x}+1} = \lim_{x\to\infty} \frac{1}{1+e^{-2x}} = \frac{1}{1+\lim_{x\to\infty} e^{-2x}} = \frac{1}{1+0} = 1$$

Since,

$$\lim_{x\to\infty} e^{-2x} = \lim_{t\to-\infty} e^t = 0$$

Exponential Functions and Their Derivatives

We can calculate the derivative of the exponential function $f(x) = b^x$ using the definition of a derivative:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{b^{x+h} - b^x}{h}$$

$$= \lim_{h \to 0} \frac{b^x b^h - b^x}{h} = \lim_{h \to 0} \frac{b^x(b^h - 1)}{h} = b^x \lim_{h \to 0} \frac{b^h - 1}{h}$$

nothing to do with x

Since we have

$$\lim_{h \to 0} \frac{(b^h - \overset{b^0}{1})}{h} = f'(0)$$ → calculation of derivative at $x=0$

We can see that if the exponential function $f(x) = b^x$ is differentiable at 0, then it is differentiable everywhere, and

$$f'(x) = f'(0)b^x$$

$f(x) = e^x$

$f'(0) = \lim_{h\to 0} \frac{f(h) - f(0)}{h-0} = \lim_{h\to 0} \frac{e^h - 1}{h}$

$= 1$

7 Definition of the Number *e*

$$e \text{ is the number such that } \lim_{h\to 0} \frac{e^h - 1}{h} = 1$$

$f'(0) = 1$

We call the function $f(x) = e^x$ the natural exponential function.

8 Derivative of the Natural Exponential Function

$$\frac{d}{dx}(e^x) = e^x$$

$\ln x$

$f^{-1}(a) = \frac{1}{f'(f^{-1}(a))}$

$b^x = e^{\ln(b^x)} = e^{x \ln b}$

Exponential Functions and Their Derivatives

Example

Differentiate the function (a) $y = e^{\tan x}$; (b) $y = e^{-4x}\sin(5x)$;

Solution:

(a) Use the Chain Rule, we let $u = tanx$, then we have $y = e^u$, so

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}.$$

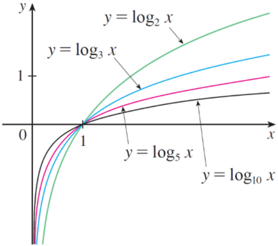
$$\frac{dy}{dx} = e^u\frac{du}{dx} = e^{\tan x}\sec^2 x.$$

(b)

$$\frac{dy}{dx} = e^{-4x}(-4)\sin(5x) + e^{-4x}\cos(5x)(5) = e^{-4x}(5\cos(5x) - 4\sin(5x)).$$

If $b > 0$ and $b \neq 1$, the exponential function $f(x) = b^x$ is either increasing or decreasing and so it is one-to-one by the Horizontal Line Test.

Therefore, it has an inverse function f^{-1}, which is called the **logarithmic function with base** b and is denoted by $\log_b x$.



$y = \log_b x = 1$
$\downarrow$
$x = b$

Use inverse function, $f^{-1}(x) = y \Leftrightarrow f(y) = x$, we have

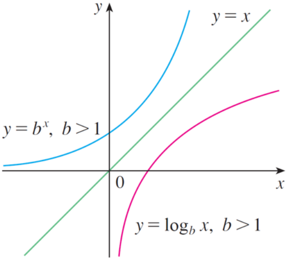
1

$$\log_b x = y \iff b^y = x$$

2

$$\log_b(b^x) = x \quad \text{for every } x \in \mathbb{R}$$
$$b^{\log_b x} = x \quad \text{for every } x > 0$$

The logarithmic function $\log_b$ has domain $(0, \infty)$. It is continuous since it is the inverse of exponential function (which is continuous).
In fact, $y = b^x$ is a very rapidly increasing function for $x > 0$, as $y = \log_b x$ is a very slowly increasing function for $x > 1$.



Logarithmic Functions

The following theorem summarizes the properties of logarithmic functions.

3 Theorem If $b > 1$, the function $f(x) = \log_b x$ is a one-to-one, continuous, increasing function with domain $(0, \infty)$ and range $\mathbb{R}$. If $x, y > 0$ and r is any real number, then

1. $\log_b(xy) = \log_b x + \log_b y$
2. $\log_b\left(\dfrac{x}{y}\right) = \log_b x - \log_b y$
3. $\log_b(x^r) = r \log_b x$

4 If $b > 1$, then

$$\lim_{x\to\infty} \log_b x = \infty \qquad \text{and} \qquad \lim_{x\to 0^+} \log_b x = -\infty$$

from positive side!.

$x=0$ is the vertical asymptote

Logarithmic Functions

Example

Find (a)$\log_4 2 + \log_4 32$; (b)$\log_2 80 - \log_2 5$; (c)$\lim_{x\to 0} \log_{10} \tan^2(x)$.

Solution:

(a) We have

$$\log_4 2 + \log_4 32 = \log_4(2 \cdot 32) = \log_4 64 = 3;$$

(b) We have

$$\log_2 80 - \log_2 5 = \log_2 \frac{80}{5} = \log_2 16 = 4;$$

→always >0

(c) As $x \to 0$, we know that $t = \tan^2(x) \to \tan^2(0) = 0$ and the values of t are positive. When $b = 10 > 1$, we have

$$\lim_{x\to 0} \log_{10} \tan^2(x) = \lim_{t\to 0^+} \log_{10} t = -\infty$$

$\lim_{x\to 0} \log_{10} \tan x$ DNE

↓

left handed

on is

not well

defined

Natural Logarithms

The logarithm with base e is called the natural logarithm.

$$\log_e x = \ln x \quad \textit{and} \quad \ln x = y \Leftrightarrow e^y = x$$

6

$$\ln(e^x) = x \qquad x \in \mathbb{R}$$
$$e^{\ln x} = x \qquad x > 0$$

$$\ln e = 1$$

7 Change of Base Formula For any positive number b ($b \neq 1$), we have

$$\log_b x = \frac{\ln x}{\ln b}$$

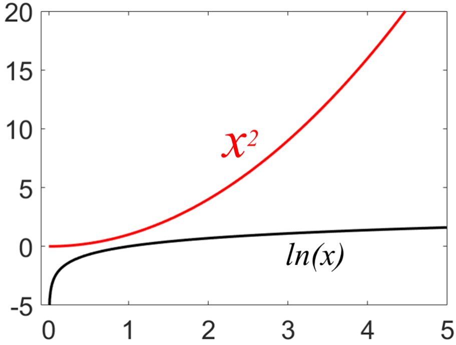
$$\lim_{x \to \infty} \ln x = \infty \qquad \lim_{x \to 0^+} \ln x = -\infty$$

In fact, we will be able to show

$\log_b x << x^p << b^x$

$$\lim_{x\to\infty} \frac{\ln x}{x^p} = 0 \quad \text{for any positive power } p.$$

So for large x, the values of lnx are very small compared with x^p.



Derivatives of Logarithmic Functions

Natural logarithmic function $y = \ln x$ is differentiable, because it is the inverse of the differentiable function $y = e^x$.

$(e^x)' = e^x$

1 $$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

$[f^{-1}(a)]' = \frac{1}{f'(f^{-1}(a))}$

Prove: Let $y = \ln x$, then $\boxed{e^y = x.}$ $y(x)$

Differentiate both sides with respect to x, we have

$e^{y(x)}$

$$e^y \frac{dy}{dx} = 1 \to \frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}$$

Further we have

2 $$\frac{d}{dx}(\ln u) = \frac{1}{u}\frac{du}{dx}$$ or $$\frac{d}{dx}[\ln g(x)] = \frac{g'(x)}{g(x)}$$

Derivatives of Logarithmic Functions

Example

Find (a) $\frac{d}{dx}\ln\sin(x)$; (b) $\frac{d}{dx}\ln\frac{x+1}{\sqrt{x-2}}$; (c) $\frac{d}{dx}\ln|x|$

Solution: (a) We have

$$\frac{d}{dx}\ln(\sin x) = \frac{1}{\sin x}\frac{d}{dx}\sin x = \frac{1}{\sin x}\cos x = \cot x$$

(b) We have

make it simpler!

$$\frac{d}{dx}\ln\frac{x+1}{\sqrt{x-2}} = \frac{d}{dx}(\ln(x+1) - \frac{1}{2}\ln(x-2))$$

$$= \frac{d}{dx}\ln(x+1) - \frac{1}{2}\frac{d}{dx}\ln(x-2) = \frac{1}{x+1} - \frac{1}{2}(\frac{1}{x-2})$$

$$(c) f(x) = \ln|x| = \begin{cases} \ln x & \forall x > 0 \\ \ln(-x) & \forall x < 0 \end{cases} \Rightarrow f'(x) = \begin{cases} \frac{1}{x} & \forall x > 0 \\ \frac{1}{-x}(-1) = \frac{1}{x} & \forall x < 0 \end{cases}$$

Thus $\frac{d}{dx}\ln|x| = \frac{1}{x}$.

Derivatives of Logarithmic Functions

6
$$\frac{d}{dx}(\log_b x) = \frac{1}{x \ln b}$$

Proof

Since $\log_b x = \frac{\ln x}{\ln b}$ and $\ln b$ is a constant, we have differentiation formula,

$$\frac{d}{dx}(log_b x) = \frac{d}{dx}\frac{lnx}{lnb} = \frac{1}{lnb}\frac{d}{dx}(lnx) = \frac{1}{xlnb}$$

For example, if we want to calculate $\frac{d}{dx}(\log_{10}(2+\sin x))$.

$$\frac{d}{dx}(\log_{10}(2+\sin x)) = \frac{1}{\ln 10}\frac{1}{(2+\sin x)}\frac{d}{dx}(2+\sin x) = \frac{\cos x}{\ln 10(2+\sin x)}$$

$\boxed{7}$

$$\frac{d}{dx}(b^x) = b^x \ln b$$

$(e^x)' = e^x \to \log_a x = \frac{1}{x \ln b} \to$ see previous slide

Proof

In fact, $e^{\ln b} = b$,

$$\frac{d}{dx}(b^x) = \frac{d}{dx}(e^{\ln b})^x = \frac{d}{dx}e^{(\ln b)x} = e^{(\ln b)x}\frac{d}{dx}((\ln b)x)$$

$$= e^{(\ln b)x}(\ln b) = b^x \ln b$$

For example, if we want to calculate $\frac{d}{dx}(10^{x^2})$.

$$\frac{d}{dx}(10^{x^2}) = 10^{x^2}(\ln 10)\frac{d}{dx}(x^2) = (2\ln 10)x10^{x^2}$$

$10^{x^2} = (e^{\ln 10})^{x^2} = e^{x^2 \ln 10}$

Derivatives of Logarithmic Functions

Example

Differentiate
$$y = \frac{x^{3/4}\sqrt{x^2+1}}{(3x+2)^5}$$

Solution:
We take logarithms of both sides of the equation and use the properties of Logarithms to simplify:

$$\ln y = \frac{3}{4}\ln x + \frac{1}{2}\ln(x^2+1) - 5\ln(3x+2)$$

Differentiating implicitly with respect to x gives

$$\frac{1}{y}\frac{dy}{dx} = \frac{3}{4}\frac{1}{x} + \frac{1}{2}\frac{2x}{x^2+1} - 5\frac{3}{3x+2}$$

Solving for $\frac{dy}{dx}$ we get,

$$\frac{dy}{dx} = y\left(\frac{3}{4x} + \frac{x}{x^2+1} - \frac{15}{3x+2}\right)$$

Because we have an explicit expression for y, we can substitute and write

$$\frac{dy}{dx} = \frac{x^{3/4}\sqrt{x^2+1}}{(3x+2)^5}\left(\frac{3}{4x} + \frac{x}{x^2+1} - \frac{15}{3x+2}\right)$$

→ don't need to try and simplify further

The calculation of derivatives of complicated functions involving products, quotients, or powers can often be simplified by taking logarithms.

The method used here is called **logarithmic differentiation**.

Example

Find the derivative of the function (a) $y = x^{\sin x}$; (b) $y = (\sin x)^{\ln x}$;
(c) $y = \sqrt{x}e^{x^2-x}(x+1)^{2/3}$

Solution: (a) we have $\ln y = \sin x \ln x$, then take the derivative on both sides with respect to x,

$$\frac{y'}{y} = \cos x \ln x + \sin x \frac{1}{x} \Rightarrow y' = y(\cos x \ln x + \frac{\sin x}{x}) = x^{\sin x}(\cos x \ln x + \frac{\sin x}{x})$$

(b) we have $\ln y = \ln x \ln(\sin x)$, then take the derivative on the both sides with respect to x,

$$\frac{y'}{y} = \frac{\ln(\sin x)}{x} + \frac{\ln x \cos x}{\sin x} \Rightarrow y' = (\sin x)^{\ln x}(\frac{\ln(\sin x)}{x} + \frac{\ln x \cos x}{\sin x})$$

Derivatives of Logarithmic Functions

(c) we have

$$\ln y = \frac{1}{2}\ln x + (x^2 - x) + \frac{2}{3}\ln(x+1).$$

We then take the derivative on the both sides with respect to x,

$$\frac{y'}{y} = \frac{1}{2x} + 2x - 1 + \frac{2}{3(x+1)}$$

Then we bring y into the equation

$$y' = y(\frac{1}{2x} + 2x - 1 + \frac{2}{3(x+1)})$$

$$y' = \sqrt{x}e^{x^2-x}(x+1)^{2/3}(\frac{1}{2x} + 2x - 1 + \frac{2}{3(x+1)})$$

Steps in Logarithmic Differentiation

1. Take natural logarithms of both sides of an equation $y = f(x)$ and use the properties of logarithms to simplify.
2. Differentiate implicitly with respect to x.
3. Solve the resulting equation for y'.

The Power Rule If n is any real number and $f(x) = x^n$, then

$$f'(x) = nx^{n-1}$$

$$x^{\alpha} = e^{\ln(x^{\alpha})} = e^{\alpha \ln x}$$

$$(x^{\alpha})' = (e^{\alpha \ln x})' = e^{\alpha \ln x} \cdot \alpha \cdot \frac{1}{x} = x^{\alpha} \cdot \alpha \cdot \frac{1}{x} = \alpha x^{\alpha - 1}$$

Derivatives of Logarithmic Functions

If $f(x) = \ln x$, then $f'(x) = 1/x$. Thus $f'(1) = 1$. We now use this fact to express the number e as a limit.

From the definition of a derivative as a limit, we have

$$f'(1) = \lim_{h\to 0} \frac{f(1+h) - f(1)}{h} = \lim_{x\to 0} \frac{f(1+x) - f(1)}{x} = \lim_{x\to 0} \frac{\ln(1+x) - \overset{0}{\cancel{\ln 1}}}{x}$$

$$= \lim_{x\to 0} \frac{1}{x} \ln(1+x) = \lim_{x\to 0} \ln(1+x)^{\frac{1}{x}}$$

Because $f'(1) = 1$, we have,

$$\lim_{x\to 0} \ln(1+x)^{\frac{1}{x}} = 1$$

Then, by the continuity of the exponential function, we have

move limit out because exponential function is continuous

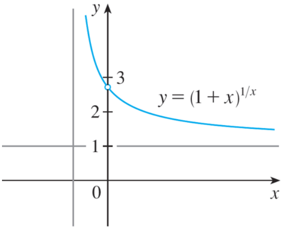
$$e = e^1 = e^{\lim_{x\to 0} \ln(1+x)^{\frac{1}{x}}} = \lim_{x\to 0} e^{\ln(1+x)^{\frac{1}{x}}} = \lim_{x\to 0} (1+x)^{\frac{1}{x}}$$

8

$$\boxed{e = \lim_{x\to 0} (1 + x)^{1/x}}$$

Derivatives of Logarithmic Functions

The function $y = (1+x)^{1/x}$ is illustrated in the figure below. Table list its values for small values of x. We can see $e \approx 2.7182818$



x	$(1+x)^{1/x}$
0.1	2.59374246
0.01	2.70481383
0.001	2.71692393
0.0001	2.71814593
0.00001	2.71826824
0.000001	2.71828047
0.0000001	2.71828169
0.00000001	2.71828181

If we replace x by $n = 1/x$, then $n \to \infty$ as $x \to 0^+$ and so an alternative expression for e is $x = \frac{1}{n}$

compound interest

9
$$\boxed{e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n}$$